Technical Report – AWS Infrastructure Implementation with EC2, VPC, and SSH Access

1. Introduction

This report aims to describe the creation and configuration of an infrastructure environment on Amazon Web Services (AWS), using fundamental resources for hosting cloud-based applications. During the lab development, networking, security, and computing resources were created and configured, along with connectivity and remote access testing via SSH.

The objective of the environment was to provide an EC2 instance accessible from the internet while following best practices for configuring VPCs, Subnets, Security Groups, and Internet Gateways.

2. Resources Created and Their Functions

2.1 Virtual Private Cloud (VPC)

The VPC named **vpc-lab** was created to provide an isolated networking environment within AWS. A VPC functions as a virtual private network where all other infrastructure resources are deployed.

The primary goal was to enable control over IP addressing, routing, and application security rules.

The following CIDR block was used:

10.0.0.0/16

This allows the creation of multiple subnets and supports future infrastructure expansion.

2.2 Subnet

A subnet named **my-subnet-01** was created using the following CIDR block:

10.0.1.0/24

This subnet represents a subdivision of the VPC and is used to organize network resources.

The EC2 instance was deployed within this subnet and received a private IP address from this range.

Additionally, the subnet was configured to automatically assign public IP addresses, enabling external access to the instance.

2.3 Internet Gateway

The resource **my-internet-gateway** was created and attached to the VPC.

The Internet Gateway is responsible for enabling communication between AWS resources and the public internet.

Without this resource, the EC2 instance would remain isolated, making SSH connections and access to external services impossible.

2.4 Route Table

A route table named **my-route-table** was created.

Its purpose is to determine how network packets are routed within the VPC.

The following route was added:

0.0.0.0/0 → Internet Gateway

This configuration ensures that all internet-bound traffic is directed to the Internet Gateway, allowing the instance to communicate externally.

2.5 Security Group

A Security Group named **MyWebServerGroup** was created.

This resource acts as a virtual firewall that controls inbound and outbound traffic to the instance.

The following inbound rule was added:

- Type: SSH
- Protocol: TCP
- Port: 22
- Source: Local computer's public IP address

This configuration enabled secure remote access to the EC2 instance.

2.6 EC2 Instance

An EC2 instance named **MyWebServer** was created.

Main characteristics:

- Operating System: Amazon Linux 2023
- Instance Type: t2.micro
- Region: us-east-1
- Subnet: my-subnet-01

The instance serves as a virtual server used to host applications, execute administrative commands, and provide a foundation for future DevOps implementations.

2.7 Key Pair

A Key Pair named **Key-Irede** was used.

This resource is responsible for secure SSH authentication.

When connecting to the instance, the locally stored private key is validated against the public key stored on the server.

This mechanism eliminates the need for password-based authentication, significantly enhancing the security of the environment.

3. Tests Performed

After the infrastructure was created, several tests were performed to validate instance connectivity.

These included:

- Verification of EC2 Status Checks
- TCP connectivity tests on port 22
- Validation of Security Group rules
- Verification of VPC routing configurations
- Internet Gateway association validation
- SSH authentication testing

Remote access was successfully established using the following command:

ssh -i Key-Irede.pem ec2-user@34.229.172.113

The login process was completed successfully, and the Amazon Linux 2023 operating system was accessed without restrictions.

4. Challenges Encountered

During the implementation of the lab, several issues were encountered and had to be investigated and resolved.

The first issue occurred during the attempt to access the EC2 instance via SSH.

Initially, the error message **"Connection Timed Out"** was displayed. After analyzing the infrastructure, it was identified that the Security Group did not contain inbound rules allowing SSH traffic on port 22.

After correcting the firewall rules, the error changed to **"Permission Denied (publickey)"**, indicating that communication with the server was functioning properly, but authentication was not being accepted.

Further investigation revealed that the original instance had been created without an associated Key Pair. Since AWS does not allow a Key Pair to be attached to an existing instance after creation, it was necessary to terminate the instance and create a new one using the correct Key Pair.

5. Conclusion

This activity provided practical insight into the operation of AWS's core infrastructure components. Networking, security, and computing resources were configured, and remote SSH access was successfully validated.

Throughout the process, challenges related to connectivity and authentication were encountered, requiring detailed infrastructure analysis and the application of troubleshooting best practices.

At the conclusion of the lab, a fully functional EC2 instance was successfully deployed, accessible from the internet and protected by appropriate security mechanisms. This experience reinforced essential knowledge for working in Cloud Computing and DevOps environments.